



Virus Hunter 6 (VH6)

A new 20-minute COVID test, claimed to be 99 percent accurate is being trialled at Surrey University. <http://digit.in/feb21-09>



Saving Grace

Study shows that people with high Omega-3 blood levels are less likely to die from COVID-19. <http://digit.in/feb21-10>

Fighting The pandemic

PART VI

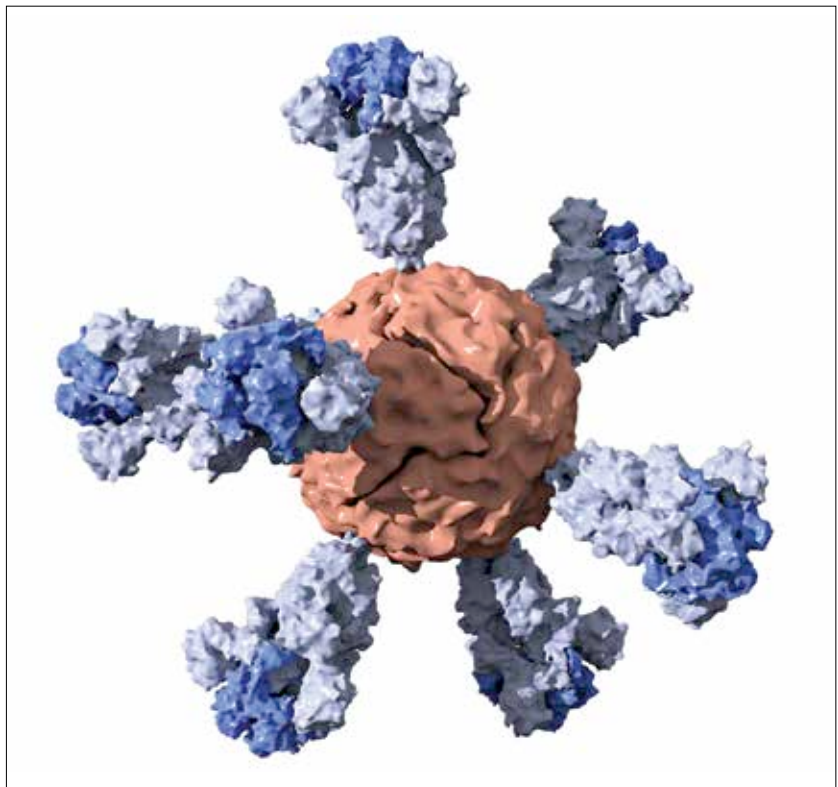
The research directions to stop the spread of the novel coronavirus

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Stanford University biochemistry lab headed by Peter S Kim had been researching nano-

particle vaccines for HIV and ebola when the pandemic hit. Although coronaviruses were not a speciality of the lab, the researchers began to work on a vaccine candidate. The nanoparticle vaccine works by using a ferritin core, which is a protein compound in the blood that contains iron, and sticking shortened SARS-CoV-2 spikes around the core. The actual coronavirus spikes are large and so the scientists looked for various approaches to shrink it, in order to make them easier to manipulate. The various modifications tested included a full spike, a spike shortened at the base, and using only the portion of the spike that binds to the cells during infection. After testing on rats, the researchers found that the nanoparticles with the shortened spikes evoked the strongest immune response. Such a vaccine could have many advantages. They are cheaper to produce at scale than nucleic acid



A schematic of a ferritin nanoparticle with shortened coronavirus spike proteins

Image: Duo Xu/Stanford

vaccines, do not need to be refrigerated, can be stored in a dried form, do not need a cold supply chain, and need only a single dose to work.

A GENERIC NANOPARTICLE VACCINE

The vaccines were originally developed for countries that might have had trouble accessing other SARS-CoV-2 vaccines, but at the present time it does not look like any country will run into such a shortage of supply. What is so interesting about this particular vaccine candidate is that the team is pivoting once more, and undertaking the even more ambi-

tious project of developing a generic coronavirus vaccine.

Researchers from Caltech are also working on a nanoparticle vaccine, one that uses bits and pieces of the spikes of different coronaviruses, into the spike structures on the nanoparticle. These spike proteins were sourced from coronavirus strains in animal populations that have the potential to make the jump to humans and turn zoonotic in the future. If and when that happens, a nanoparticle vaccine designed to prevent their spread can prevent a future pandemic from taking hold. The vaccine platform is called a mosaic